

003 — SSH Cheatsheet

 SSH (Secure Shell) is how you securely connect to remote servers and authenticate with GitHub. This cheatsheet covers everything you need for the bootcamp.

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1. SSH Key Generation

1.1 Understanding SSH Keys

SSH uses a **key pair**:

- **Private key** (`~/.ssh/id_ed25519`) — stays on YOUR machine. Never share this.
- **Public key** (`~/.ssh/id_ed25519.pub`) — goes on remote servers/GitHub. Safe to share.

Think of it like a lock and key:

- The **public key** is the lock (you put it on doors you want to open)
- The **private key** is the key (only you have it)

1.2 Generate a Key (All Platforms)

Works identically on **macOS Terminal**, **WSL Ubuntu**, and **Windows PowerShell**.

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

Prompts:

```
Generating public/private ed25519 key pair.  
Enter file in which to save the key (/home/user/.ssh/id_ed25519): [Press  
Enter]  
Enter passphrase (empty for no passphrase): [Press Enter or set passphrase]  
Enter same passphrase again: [Press Enter]
```

Output:

```
Your identification has been saved in /home/user/.ssh/id_ed25519
Your public key has been saved in /home/user/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:abc123... your_email@example.com
```

1.3 Start SSH Agent & Add Key

macOS / WSL (Linux):

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

Windows PowerShell (native, if not using WSL):

```
# Start the ssh-agent service
Get-Service ssh-agent | Set-Service -StartupType Automatic
Start-Service ssh-agent

# Add your key
ssh-add $env:USERPROFILE\.ssh\id_ed25519
```

1.4 View Your Public Key

macOS / WSL:

```
cat ~/.ssh/id_ed25519.pub
```

Windows PowerShell:

```
Get-Content $env:USERPROFILE\.ssh\id_ed25519.pub
```

Example output:

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHx7...long_string...abc
your_email@example.com
```

1.5 Check Existing Keys

Before generating, check if you already have keys:

```
ls -la ~/.ssh/
```

Look for files like:

- `id_ed25519` / `id_ed25519.pub` (Ed25519 — recommended)
- `id_rsa` / `id_rsa.pub` (RSA — older but still works)

If you already have a key pair, you can reuse them.

2. Adding SSH Key to GitHub

2.1 Copy Your Public Key

macOS (copy to clipboard):

```
pbcopy < ~/.ssh/id_ed25519.pub
```

WSL/Linux (copy to clipboard):

```
cat ~/.ssh/id_ed25519.pub | clip.exe
```

Or just display and manually select+copy:

```
cat ~/.ssh/id_ed25519.pub
```

2.2 Add to GitHub

1. Go to: <https://github.com/settings/ssh/new>
2. **Title:** `Bootcamp Laptop` (or any descriptive name)
3. **Key type:** `Authentication Key`
4. **Key:** Paste your public key
5. Click **Add SSH key**

2.3 Test GitHub Connection

```
ssh -T git@github.com
```

 **Success:**

```
Hi YOUR_USERNAME! You've successfully authenticated, but GitHub does not
provide shell access.
```

✗ Failure:

```
Permission denied (publickey).
```

→ See [Troubleshooting Section 6.1](#)

2.4 Clone Using SSH (not HTTPS)

Once your key is set up, always clone using SSH URLs:

```
# ✔ SSH (use this)
git clone git@github.com:proxeon/bootcamp.git

# ✗ HTTPS (avoid – requires password/token each time)
git clone https://github.com/proxeon/bootcamp.git
```

3. Adding SSH Key to DigitalOcean VPS

3.1 Method A: Add Key During Droplet Creation (Recommended)

When creating a new Droplet on DigitalOcean:

1. In the **Authentication** section, select **SSH Keys**
2. Click **New SSH Key**
3. Paste your public key (`cat ~/.ssh/id_ed25519.pub`)
4. Give it a name (e.g., `My Laptop`)
5. Click **Add SSH Key**
6. Continue creating the Droplet

Now you can immediately SSH into the server:

```
ssh root@YOUR_DROPLET_IP
```

3.2 Method B: Add Key After Droplet Creation

If you already created a Droplet with password auth:

From your local machine:

```
ssh-copy-id root@YOUR_DROPLET_IP
```

You'll be prompted for the root password (the one DigitalOcean emailed you), then your key will be installed.

Or manually:

```
# SSH in with password first
ssh root@YOUR_DROPLET_IP

# On the VPS, create .ssh directory
mkdir -p ~/.ssh
chmod 700 ~/.ssh

# Add your public key (paste the content of id_ed25519.pub)
echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5... your_email" >>
~/.ssh/authorized_keys
chmod 600 ~/.ssh/authorized_keys
```

3.3 Test VPS Connection

```
ssh root@YOUR_DROPLET_IP
```

✅ **Success:** You see the server welcome message and a `root@droplet:~#` prompt.

First-time connection — you'll see:

```
The authenticity of host '123.456.78.90' can't be established.
ED25519 key fingerprint is SHA256:abc123...
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Type `yes` and press Enter. This is normal for first connections.

3.4 Add Key to DigitalOcean Account (for future Droplets)

1. Go to: <https://cloud.digitalocean.com/account/security>
2. Scroll to **SSH Keys**
3. Click **Add SSH Key**
4. Paste your public key and name it

Now every new Droplet you create will automatically include this key.

4. SSH Config File

The SSH config file (`~/.ssh/config`) lets you create shortcuts for SSH connections. Instead of typing `ssh root@167.71.123.45`, you can just type `ssh vps`.

4.1 Create/Edit the Config File

```
nano ~/.ssh/config
```

Or with any editor:

```
code ~/.ssh/config    # VS Code
vim ~/.ssh/config     # Vim
```

4.2 Basic Config Structure

```
# ——— GitHub ———
Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519
  AddKeysToAgent yes

# ——— DigitalOcean VPS ———
Host vps
  HostName 167.71.123.45
  User root
  IdentityFile ~/.ssh/id_ed25519
  AddKeysToAgent yes

# ——— Alternative: VPS with custom port ———
Host vps-secure
  HostName 167.71.123.45
  User root
  Port 2222
  IdentityFile ~/.ssh/id_ed25519
```

4.3 Set Correct Permissions

```
chmod 600 ~/.ssh/config
```

4.4 Usage After Config

```
# Before config:
ssh root@167.71.123.45
```

```
# After config:
ssh vps
```

```
# SCP files using alias
scp ./file.txt vps:/root/file.txt

# Run remote command
ssh vps "docker ps"
```

4.5 Multiple Keys for Different Services

If you have separate keys for GitHub and your VPS:

```
# GitHub uses one key
Host github.com
    HostName github.com
    User git
    IdentityFile ~/.ssh/id_ed25519_github

# VPS uses a different key
Host vps
    HostName 167.71.123.45
    User root
    IdentityFile ~/.ssh/id_ed25519_vps
```

4.6 Wildcard Config (apply to all hosts)

```
# Apply to all connections
Host *
    AddKeysToAgent yes
    IdentityFile ~/.ssh/id_ed25519
    ServerAliveInterval 60
    ServerAliveCountMax 3
```

ServerAliveInterval prevents SSH from disconnecting due to inactivity.

5. Common SSH Commands

5.1 Connecting

Command	Description
<code>ssh root@IP</code>	Connect to server as root

Command	Description
<code>ssh user@IP</code>	Connect as specific user
<code>ssh -p 2222 root@IP</code>	Connect on custom port
<code>ssh vps</code>	Connect using config alias
<code>ssh -v root@IP</code>	Verbose mode (for debugging)
<code>ssh -vvv root@IP</code>	Extra verbose (maximum debugging)

5.2 Key Management

Command	Description
<code>ssh-keygen -t ed25519 -C "email"</code>	Generate new key pair
<code>ssh-keygen -t ed25519 -f ~/.ssh/mykey</code>	Generate with custom filename
<code>ssh-add ~/.ssh/id_ed25519</code>	Add key to agent
<code>ssh-add -l</code>	List keys in agent
<code>ssh-add -D</code>	Remove all keys from agent
<code>ssh-keygen -R hostname</code>	Remove host from known_hosts
<code>ssh-keygen -l -f ~/.ssh/id_ed25519.pub</code>	Show key fingerprint

5.3 File Transfer (SCP)

Command	Description
<code>scp file.txt root@IP:/path/</code>	Upload file to server
<code>scp root@IP:/path/file.txt .</code>	Download file from server
<code>scp -r folder/ root@IP:/path/</code>	Upload folder recursively
<code>scp -P 2222 file.txt root@IP:/path/</code>	SCP with custom port

5.4 Remote Command Execution

```
# Run single command on remote server
ssh vps "docker ps"

# Run multiple commands
ssh vps "cd ~/kanban-prod && docker compose ps"

# Run script remotely
ssh vps 'bash -s' < local-script.sh
```

5.5 SSH Tunneling (Port Forwarding)


```
# Forward local port 8080 to remote server's port 80
ssh -L 8080:localhost:80 root@IP

# Access remote database locally (e.g., PostgreSQL on VPS)
ssh -L 5433:localhost:5432 root@IP
# Then connect to localhost:5433 with your local DB client

# Reverse tunnel: make your local port 3000 accessible on the VPS
ssh -R 9000:localhost:3000 root@IP
```

6. Troubleshooting

6.1 "Permission denied (publickey)"

Diagnosis:

```
ssh -vT git@github.com
```

Look for lines like:

```
debug1: Offering public key: /home/user/.ssh/id_ed25519
debug1: Server accepts key: /home/user/.ssh/id_ed25519
```

Common causes & fixes:

Cause	Fix
Key not added to agent	<code>ssh-add ~/.ssh/id_ed25519</code>
Wrong key on GitHub	Re-check <code>cat ~/.ssh/id_ed25519.pub</code> matches GitHub
SSH agent not running	<code>eval "\$(ssh-agent -s)"</code> then re-add key
Wrong file permissions	See Section 6.5
Multiple keys, wrong one offered	Use SSH config with explicit <code>IdentityFile</code>

6.2 "Host key verification failed"

```
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@ WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED! @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
```

Cause: The server's fingerprint changed (e.g., you recreated the Droplet).

Fix:

```
# Remove the old fingerprint
ssh-keygen -R YOUR_DROPLET_IP

# Or remove specific line from known_hosts
nano ~/.ssh/known_hosts
# Delete the line containing the IP address
```

Then try connecting again — you'll be asked to accept the new fingerprint.

6.3 "Connection refused" or "Connection timed out"

Error	Likely Cause	Fix
Connection refused	SSH not running on server	Check server status on DigitalOcean dashboard
Connection refused	Wrong port	Verify port (default 22): <code>ssh -p 22 root@IP</code>
Connection timed out	Firewall blocking	Check VPS firewall: <code>ufw status</code>
Connection timed out	Wrong IP address	Verify IP on DigitalOcean dashboard
Connection timed out	Server is down	Reboot via DigitalOcean console

6.4 "Too many authentication failures"

Cause: SSH agent is trying multiple keys and failing.

Fix:

```
# Specify exact key to use
ssh -i ~/.ssh/id_ed25519 root@IP

# Or clear agent and add only the right key
ssh-add -D
ssh-add ~/.ssh/id_ed25519
```

6.5 File Permission Issues

SSH is strict about file permissions. Fix with:

```
# Fix .ssh directory
chmod 700 ~/.ssh

# Fix private key
chmod 600 ~/.ssh/id_ed25519

# Fix public key
```

```
chmod 644 ~/.ssh/id_ed25519.pub

# Fix config file
chmod 600 ~/.ssh/config

# Fix authorized_keys (on server)
chmod 600 ~/.ssh/authorized_keys

# Fix ownership (if needed)
chown -R $USER:$USER ~/.ssh
```

If permissions are wrong, SSH will silently refuse to use the key.

6.6 WSL-Specific Issues

Problem: SSH key generated in Windows is not accessible in WSL (or vice versa).

Fix: Generate and use keys **inside WSL only**. Windows and WSL have separate filesystems.

```
# Check where you are
pwd
# Should be /home/yourname (WSL) not /mnt/c/Users/... (Windows)

# Your WSL SSH directory
ls ~/.ssh/
# Should show id_ed25519 and id_ed25519.pub
```

⚠ Never store SSH keys on the Windows filesystem (`/mnt/c/...`) and use them from WSL. The file permissions will be wrong and SSH will ignore them.

6.7 "Agent has no identities" or Key Not Persisting

Problem: SSH key disappears after terminal restart.

Fix for macOS: Add to `~/.ssh/config`:

```
Host *
  UseKeychain yes
  AddKeysToAgent yes
  IdentityFile ~/.ssh/id_ed25519
```

Fix for WSL/Linux: Add to `~/.bashrc`:

```
# Auto-start SSH agent
if [ -z "$SSH_AUTH_SOCK" ]; then
  eval "$(ssh-agent -s)" > /dev/null 2>&1
```

```
ssh-add ~/.ssh/id_ed25519 > /dev/null 2>&1  
fi
```

Then reload:

```
source ~/.bashrc
```

6.8 Testing Connection Step by Step

When something isn't working, debug systematically:

```
# Step 1: Check if key exists  
ls -la ~/.ssh/id_ed25519  
  
# Step 2: Check permissions  
stat -c "%a %U" ~/.ssh/id_ed25519  
# Should show: 600 yourusername  
  
# Step 3: Check if agent is running  
echo $SSH_AUTH_SOCK  
# Should show a path, not empty  
  
# Step 4: Check if key is loaded in agent  
ssh-add -l  
# Should show your key fingerprint  
  
# Step 5: Verbose connection test  
ssh -vvv root@YOUR_IP 2>&1 | head -50  
  
# Step 6: Check if port is reachable  
nc -zv YOUR_IP 22  
# Should show: Connection to YOUR_IP 22 port [tcp/ssh] succeeded!
```

7. Quick Reference Card

Key Generation Cheat Sheet

```
# Generate key  
ssh-keygen -t ed25519 -C "email@example.com"  
  
# Start agent  
eval "$(ssh-agent -s)"  
  
# Add key to agent  
ssh-add ~/.ssh/id_ed25519
```

```
# View public key (copy this to GitHub/VPS)
cat ~/.ssh/id_ed25519.pub

# Test GitHub
ssh -T git@github.com

# Test VPS
ssh root@YOUR_DROPLET_IP
```

SSH Config Template

```
# ~/.ssh/config

Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519

Host vps
  HostName YOUR_DROPLET_IP
  User root
  IdentityFile ~/.ssh/id_ed25519
  ServerAliveInterval 60
```

Permissions Cheat Sheet

```
~/.ssh/           → 700 (drwx-----)
~/.ssh/id_ed25519 → 600 (-rw-----)
~/.ssh/id_ed25519.pub → 644 (-rw-r--r--)
~/.ssh/config     → 600 (-rw-----)
~/.ssh/known_hosts → 644 (-rw-r--r--)
~/.ssh/authorized_keys → 600 (-rw-----) [on server]
```

Emergency Fixes

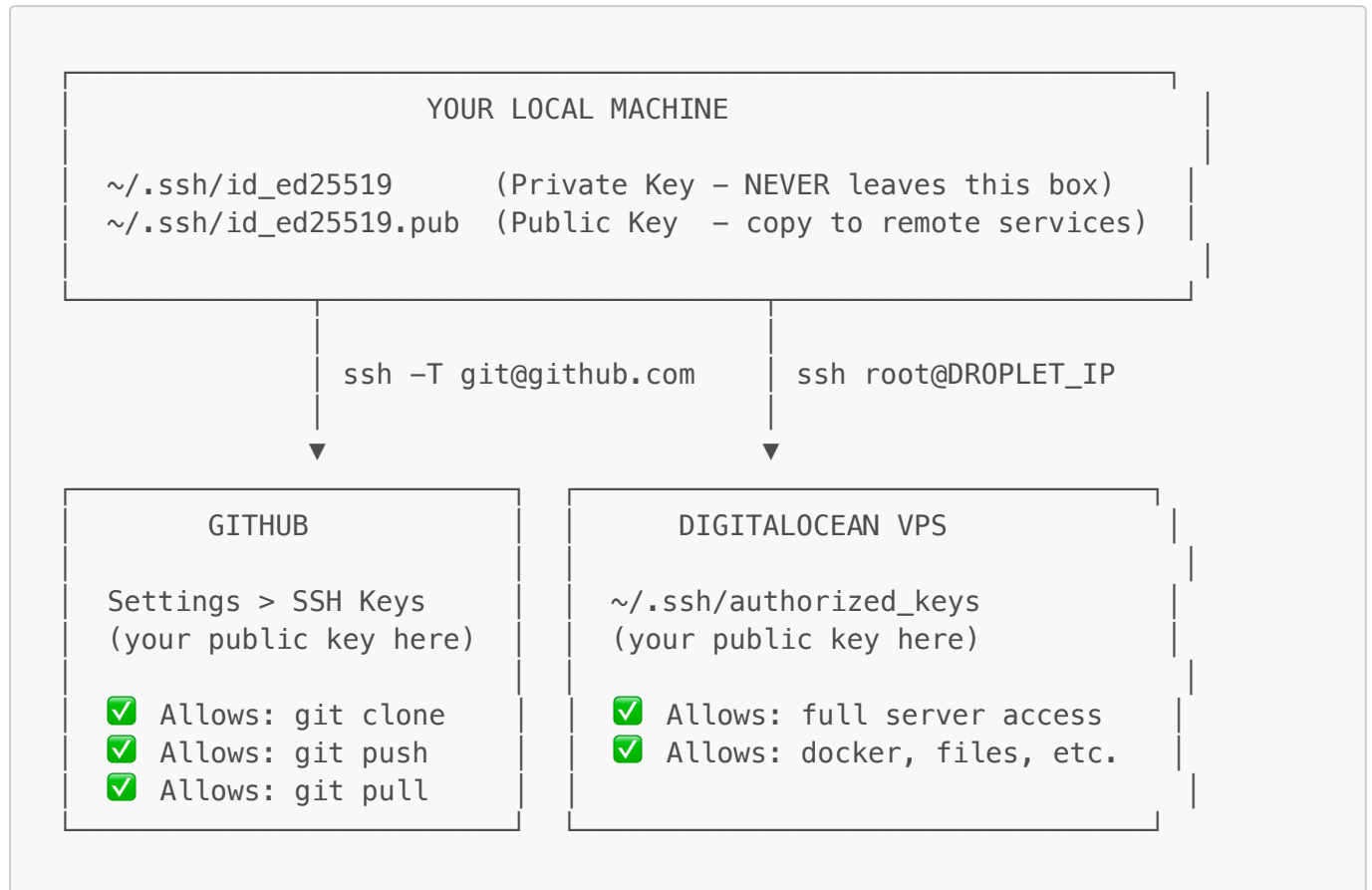
```
# "I can't connect to anything"
eval "$(ssh-agent -s)" && ssh-add ~/.ssh/id_ed25519

# "My known_hosts is messed up"
ssh-keygen -R HOSTNAME_OR_IP

# "Permission denied on everything"
chmod 700 ~/.ssh && chmod 600 ~/.ssh/id_ed25519 && chmod 644
~/.ssh/id_ed25519.pub
```

```
# "I need to start over"
rm -rf ~/.ssh && ssh-keygen -t ed25519 -C "email@example.com"
```

Diagram: SSH Flow



Bootcamp Day: Quick Setup Summary

If you followed the [Pre-Bootcamp Setup Guide \(000\)](#), you should already have:

- ✓ SSH key generated
- ✓ Key added to GitHub
- ✓ `ssh -T git@github.com` works

On bootcamp day, you'll additionally:

1. Create a DigitalOcean Droplet (add your SSH key during creation)
2. Add VPS entry to your `~/.ssh/config`
3. SSH into VPS and deploy the application

```
# After creating Droplet, add to config:
echo "
Host vps
  HostName YOUR_NEW_DROPLET_IP
  User root
  IdentityFile ~/.ssh/id_ed25519
```

```
    ServerAliveInterval 60  
" >> ~/.ssh/config
```

```
# Then simply:  
ssh vps
```